

AMALJOSH MAADHAV J

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Profile / Career Objective

Applied AI and Data Science undergraduate, building end-to-end machine learning systems from data preprocessing to deployment. Experienced in Computer Vision, NLP, and predictive analytics focused on real-world problem solving and scalable AI systems. Interested in developing production-ready intelligent applications.

Education

Saveetha Engineering College, Chennai

B.Tech in Artificial Intelligence and Data Science

2023 – 2027

Chennai

Dheeran Chinnamalai International Residential School

Higher Secondary (CBSE)

2021 – 2023

Krishnagiri

Experience

KaviTamil Solutions

July 2024 – August 2024

Artificial Intelligence Trainee

Chennai

- Applied supervised and unsupervised ML algorithms on real datasets using Python and Scikit-learn.
- Performed preprocessing, feature engineering, model evaluation and optimization in Kaggle projects.
- Implemented end-to-end ML workflow from problem definition to deployable solution.

Achievements

Semi-Finalist, Flipkart GRID 6.0 Robotics Challenge — Built an autonomous object detection and sorting system, competing among 1L+ registrations nationwide.

Finalist, Nasscom Automation Using Agentic AI — Developed an agentic AI-based workflow automation solution, selected among top teams nationally.

Projects

FlipVision – Intelligent E-Commerce Platform with AI Vision | *YOLO, PaddleOCR, ResNet50, SQLite* **Nov 2024**

- Built multi-model computer vision data pipeline integrating YOLO, PaddleOCR, and ResNet50 to automate brand detection, freshness scoring, and expiry date extraction from product images.
- Designed structured Django ORM data model with SQLite (PostgreSQL-ready) supporting end-to-end e-commerce workflows including product management, order tracking, and multi-role user data.

Reva AI – Unified Recruitment Management System | *PaddleOCR, YOLO, Firebase, MongoDB*

Oct 2025

- Built document intelligence data pipeline using YOLO and PaddleOCR to automatically extract and structure candidate data from government identity forms into MongoDB.
- Deployed scalable REST API services (Node.js + MongoDB) integrating multilingual NLP and speech processing to support automated candidate processing workflows.

MatExtractAI – LLM Based Research PDF Extraction | *Python, PyMuPDF, Camelot, Pandas, NLP*

Jan 2026

- Designed offline multi-agent LLM pipeline to parse and transform research PDFs into structured, evidence-linked datasets using PyMuPDF and Camelot for table and text extraction.
- Implemented cross-agent validation and semantic consistency checks to ensure data quality, enabling reproducible materials informatics analysis pipelines.

TRIAGEON – ML Health Triage System | *Python, Scikit-learn, React, REST APIs, MongoDB, Docker*

Feb 2026

- Developed ML-driven clinical triage platform combining multi-disease prediction models with explainable urgency scoring, integrating Python ML APIs with Node.js backend and MongoDB data store.
- Containerized microservices using Docker enabling real-time patient risk assessment with reliable data flow across distributed components.

TraceGuard AI – Offline Cyber Investigation Assistant | *Python, FastAPI, FAISS, RAG*

Mar 2026

- Built offline forensic data pipeline ingesting and parsing Windows Event Logs (.evtx) and network traffic (.pcap), transforming raw logs into structured evidence using sentence-transformer embeddings and FAISS vector database.
- Integrated RAG pipeline with local LLM inference (Qwen2.5-3B via Ollama) to enable natural language querying over forensic datasets and auto-map findings to MITRE ATT&CK techniques.

Technical Skills

Programming: Python, C, Java

Machine Learning & Deep Learning: Regression, Classification, Clustering, CNNs, RNNs, Transformers, Feature Engineering, Model Evaluation, Imbalanced Data Handling

Libraries: NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, OpenCV, Matplotlib, Seaborn

Data and Databases: SQL, MongoDB, Data Preprocessing, EDA

AI Domains: NLP, Computer Vision, Predictive Analytics, LLMs, OCR, Speech Processing

Tools: Git, Docker, Jupyter, Google Colab, Kaggle, Tableau

AI Infrastructure: FAISS, Retrieval Augmented Generation (RAG), FastAPI

Concepts: REST APIs, Microservices, Model Deployment, Explainable AI, Cross Validation, Data Leakage Prevention